INCIDENT INVESTIGATION REPORT

Scenario 1 | Windows endpoint telemetry and Wazuh alert assessment

Item	Value
Evidence sources	Scenario_1_Archives.json; Scenario_1_Alerts.json
Archive file SHA-256	c8b1334e44abb6f21048e3df53d56ab2709ac4abcb98ba416ffdbf760f8aef19
Alerts file SHA-256	2ba5de33993dc570cf35db960312b6e014886b6d20cfc25d1ee16dee12cb80cb
Host / Wazuh agent	DESKTOP-5OG7C28 / Windows_VM (agent 001)
Agent metadata IP	192.168.232.130 (host agent address; not an external connection)
Observed time span	23 Oct 2026, 13:33:25.413 - 13:41:45.609 UTC
Primary activity interval	23 Oct 2026, 13:37:25.182 - 13:37:27.231 UTC
Disposition	Command execution observed; malicious outcome not established
Overall confidence	HIGH for the evidence-bounded reconstruction

1. Executive Summary

Observed: On Windows endpoint DESKTOP-5OG7C28, a PowerShell session belonging to **DESKTOP-5OG7C28\Darragh Casey** launched **four hostname.exe** and **two whoami.exe** processes over approximately one second. The same parent PowerShell instance then spawned a PowerShell command that checked a named batch-file path, followed by a second PowerShell command invoking **Start-Process** for the batch file. That second PowerShell instance created **cmd.exe /c** with a command line targeting **Scenario_1_script.bat**. This establishes a process-level batch-file launch attempt, not the batch file's contents, success, or consequences [Sysmon records 27940-27950].

Separate activity: Earlier Wazuh alerts for newly created EXE files trace to Microsoft Defender's MpCmdRun.exe during a signature/update command sequence [27934-27938]. Two later alerts refer to **__PSScriptPolicyTest_*.ps1** files created by PowerShell [27944-27945, 27948-27949]. Neither alert group, by itself, establishes malicious ingress or lateral movement.

Assessment: The path **C:\AtomicRedTeam\ExternalPayloads\Scenario_1_script.bat** and clustered discovery/execution pattern are consistent with purposefully orchestrated laboratory activity. This is an inference from visible naming and sequence, not verification of any specific test framework or known scenario. No supplied event demonstrates data theft, persistence, privilege escalation, network transfer, or effects of the BAT contents. The alerting signal is real, but its security meaning is narrower than some rule labels and ATT&CK tags suggest.

2. Scope of Activity

The evidence comprises **32 archive records** and **five Wazuh alert records**. Each alert refers to an event also present in the archive (matching host, channel and Windows EventRecordID), so the files represent **32 distinct Windows events, not 37**. Archive breakdown: 20 Sysmon Event ID 1 process creations, four Sysmon Event ID 11 file creations, four Security Event ID 4624 logons, and four Security Event ID 4672 privilege-assignment records.

All provided events relate to the same host and Wazuh agent; no second endpoint is represented. The user process cluster runs in a medium-integrity user context, session 1, LogonId 0xFCD57. Background service/update events run under NT AUTHORITY\SYSTEM, LogonId 0x3E7. Observed paths concern local Windows, Temp, and C:\AtomicRedTeam locations. No Sysmon network-connectivity events, file-content captures, script-content logs, process exit statuses, or remote-host records are supplied.

Interpretation therefore covers events visible *within this export*; a missing event is not evidence that the operation could not have happened elsewhere or outside the captured time span.

3. Detailed Investigation

3.1 Local host and account discovery

Observed: Between 13:37:25.182 and 13:37:26.124 UTC, four executions of C:\Windows\System32\HOSTNAME.EXE (PIDs 1572, 8868, 5276, 7596) and two of C:\Windows\System32\whoami.exe (PIDs 5856, 3892) occurred [27940-27943, 27946-27947]. All six list parent PID **1392**, parent process GUID **{4b2ce92d-5510-6adb-fd01-000000001a00}**, image **powershell.exe**, same logged-on user, medium integrity, and working directory **C:\Users\Darragh Casey\Desktop\s_1**.

Inference: These commands likely queried the host name and executing user as a brief local discovery step. The repetition and shared parent show a related process burst; the returned host name and user output are not in the evidence. No process-creation record for the parent PID 1392 appears in this export, so its original trigger cannot be determined.

3.2 Batch path check, launch and parent-child correlation

Observed: Parent PowerShell PID 1392 spawned PowerShell PID **10880** with command text containing **if (Test-Path ...Scenario_1_script.bat) {exit 0} else {exit 1}** [27944]. It then spawned PowerShell PID **4500** with **Start-Process** directed at the same batch file through a path containing **atomics\..\ExternalPayloads** [27948]. The next logged process is **cmd.exe**, PID **6000**, parent PID 4500, parent GUID exactly matching PowerShell PID 4500, with **/c** and the normalized target **C:\AtomicRedTeam\ExternalPayloads\Scenario_1_script.bat** [27950].

Inference: The check and launch refer to the same effective location because **atomics\..** resolves to the parent directory in an ordinary Windows path. The direct process-parent link between PID 4500 and PID 6000 is stronger evidence than temporal proximity alone. Creation of cmd.exe demonstrates that the launch progressed to the command interpreter; this does **not** prove what commands were in the batch file, whether they completed, or whether any intended effects occurred. The Test-Path return value is absent.

3.3 Temporary .ps1 creations and Wazuh rule 92213

Observed: PowerShell PID 10880 created **__PSScriptPolicyTest_qhypv0d4.v5c.ps1** at 13:37:25.773, and PID 4500 created **__PSScriptPolicyTest_5aj3nptp.trx.ps1** at 13:37:26.445, both in the same user Temp directory [27945, 27949]. Wazuh flagged both with severity level 15, rule 92213, text "Executable file dropped in folder commonly used by malware" and tag **T1105**.

Inference: The filename pattern and matching PowerShell creators are consistent with transient execution-policy test artifacts, not evidence of externally downloaded malicious executables. The records do not contain those files' contents; their specific purpose cannot be proven from filenames alone. The alert description overstates what the file-creation events establish.

3.4 Defender update activity and Wazuh rule 92217

Observed: Defender service **MsMpEng.exe** (parent PID 3132) spawned **MpCmdRun.exe** with **SignatureUpdate -ScheduleJob -RestrictPrivileges** [27934], **SignaturesUpdateService -ScheduleJob -UnmanagedUpdate** [27935], and **SignaturesUpdateService -ScheduleJob -HttpDownload -RestrictPrivileges** [27936]. The last instance, PID **4396**, created **mpam-bce1d755.exe** and **UpdatePlatform.exe** below **C:\Windows\SystemTemp\...MpCommU** [27937-27938]. Wazuh rule 92217, severity level 6, labelled each "Executable dropped in Windows root folder" with T1570.

Inference: The executable names, update arguments, service parent and common process GUID support a Defender update interpretation. The command text **-HttpDownload** does not independently prove an actual completed network download or a remote destination. No inter-host transfer is shown, so T1570 is unsupported despite its alert metadata. The source evidence does not cryptographically validate the two dropped files, but also offers no positive malicious indicator for them.

3.5 Other local system activity and association limits

Observed: Two SecurityHealthHost.exe launches [27932-27933], a GPSvcGroup svchost.exe start [27939], Windows Terminal OpenConsole.exe [27951], MicrosoftEdgeUpdate.exe [27952], UsoClient.exe [27953], and MoUsoCoreWorker.exe instances [27954-27955] appear in the collection. Security Event IDs 4624 and 4672 occur in four pairs [83312-83319]; 4624 records specify service **logon type 5** for **SYSTEM** initiated by services.exe, with no source network address.

Assessment: These records look compatible with local operating-system, console, and update operations; the evidence does not connect them to actions inside Scenario_1_script.bat. OpenConsole.exe appears 144 ms after the cmd.exe creation, but its parent process GUID is all zeroes, so direct process ancestry to cmd.exe is **not proven**. SYSTEM privilege logs indicate service context, not evidence that the user's batch sequence elevated privileges. The included background activity is not used to enlarge the incident narrative.

4. Evidence Analysis

Correlations use Windows event channel + EventRecordID to identify the underlying record, and Sysmon **ProcessGuid / ParentProcessGuid** for parent-child and file-creator links. Windows **systemTime / eventdata.utcTime** is preferred for chronology; Wazuh timestamps are collection times (recorded at UTC+02:00), not a second execution timeline. No external scenario reference or answer key was used. The tables below are a selective evidence index; all record references belong to **Scenario_1_Archives.json**, with the five flagged events also in **Scenario_1_Alerts.json**.

Record ID(s)	Type	Critical field(s)	Evidentiary value
27940, 27941, 27942, 27946	Sysmon 1	Four HOSTNAME.EXE launches; same PowerShell parent PID 1392 / GUID {4b2ce92d-5510-6adb-fd01-000000001a00}.	Direct evidence of hostname command executions; outputs unrecorded.
27943, 27947	Sysmon 1	Two whoami.exe launches; same parent PowerShell, same user session.	Direct evidence of identity-query command executions; output unrecorded.
27944	Sysmon 1	PowerShell PID 10880; command contains Test-Path, Scenario_1_script.bat and exit-code branches.	Command was invoked; existence-test result unknown.
27945	Sysmon 11 + alert 92213	PID 10880 created __PSScriptPolicyTest_qhypv0d4.v5c.ps1 in user Temp.	PowerShell-originated file; does not establish ingress T1105.
27948	Sysmon 1	PowerShell PID 4500; Start-Process targets atomics\..\ExternalPayloads\Scenario_1_script.bat.	Direct invocation request for the named batch file.
27949	Sysmon 11 + alert 92213	PID 4500 created __PSScriptPolicyTest_5aj3nptp.trx.ps1 in user Temp.	Same process as launch command, not evidence of external transfer.
27950	Sysmon 1 + alert 92004	cmd.exe PID 6000; /c references C:\AtomicRedTeam\ExternalPayloads\Scenario_1_script.bat; ParentProcessGuid equals PID 4500 ProcessGuid.	Confirms PowerShell-to-cmd process linkage and launch attempt.
27934-27938	Sysmon 1/11 + alert 92217	Defender update-command chain; MpCmdRun PID 4396 wrote mpam-bce1d755.exe and UpdatePlatform.exe.	Separately attributable file creations; no lateral transfer evidenced.
83312-83319	Security 4624/4672	Four SYSTEM service-logon and privilege-assignment pairs; type 5 logon via services.exe.	Not direct evidence of user privilege escalation or remote access.
27951-27955	Sysmon 1	OpenConsole.exe, Edge updater and Windows Update components; distinct parents.	Contextual activity without demonstrated batch-script linkage.

Exact process-link identifiers

Parent PowerShell (creation record absent): PID 1392; GUID `{4b2ce92d-5510-6adb-fd01-000000001a00}`.

Test-Path PowerShell: PID 10880; GUID `{4b2ce92d-6315-6adb-aa04-000000001a00}` [27944-27945].

Start-Process PowerShell: PID 4500; GUID `{4b2ce92d-6316-6adb-ae04-000000001a00}` [27948-27949].

Child cmd.exe: PID 6000; GUID `{4b2ce92d-6317-6adb-b004-000000001a00}`; ParentProcessGuid `{4b2ce92d-6316-6adb-ae04-000000001a00}` [27950].

Defender updater: PID 4396; GUID `{4b2ce92d-6233-6adb-a304-000000001a00}` [27936-27938].

Relevant command lines (normalised for readability)

In these excerpts, the JSON/Wazuh escaping is removed and path spelling is shown as represented in the Sysmon human-readable message. No recovered batch-file content is available.

PID 10880 [27944]

```
"powershell.exe" & {if (Test-Path "C:\AtomicRedTeam\atomics\..\ExternalPayloads\Scenario_1_script.bat") {exit 0} else {exit 1}}
```

PID 4500 [27948]

```
"powershell.exe" & {Start-Process "C:\AtomicRedTeam\atomics\..\ExternalPayloads\Scenario_1_script.bat"}
```

PID 6000 [27950]

```
C:\WINDOWS\system32\cmd.exe /c ""C:\AtomicRedTeam\ExternalPayloads\Scenario_1_script.bat" "
```

PID 4396 [27936]

```
MpCmdRun.exe SignaturesUpdateService -ScheduleJob -HttpDownload -RestrictPrivileges
```

Alert adjudication

Wazuh ID / Count	Rule / tag	Investigator assessment
92217 / 2	Executable in Windows root; T1570	The two EXEs were written by Defender MpCmdRun.exe in SystemTemp\...MpCommU. No second host or lateral tool movement evidenced; do not adopt T1570.
92213 / 2	File in malware-prone folder; T1105	Both records name .ps1 files with __PSScriptPolicyTest prefix created by PowerShell. No source connection or transferred payload evidenced; do not adopt T1105.
92004 / 1	PowerShell spawned command shell; T1059.003	True positive for a PowerShell-parented cmd.exe. Supported command-shell execution, not maliciousness of the batch contents.

The Wazuh severity levels (6, 15 and 4 respectively) are **rule priorities, not independent forensic proof of compromise**. The "firedtimes" counters are rule-level cumulative counters; they are not evidence of additional distinct occurrences in the uploaded records.

5. Timeline of Events

All times are 23 Oct 2026, UTC; millisecond precision comes from Sysmon eventdata.utcTime or Windows Security systemTime. Related events are consolidated only where their individual record IDs remain explicit.

UTC	Record(s)	Event / relationship
13:33:25.413-.628	27932-27933	Two SecurityHealthHost.exe instances start under the interactive user; no established relationship to the later batch launch.
13:33:26.333-.616	27934-27935; 83312-83313	Defender MpCmdRun.exe update tasks run as SYSTEM; SYSTEM service logon/privilege audit pair.
13:33:39.753	27936	Defender MpCmdRun.exe PID 4396 starts with SignaturesUpdateService, -HttpDownload and -RestrictPrivileges.
13:33:39.880-.943	27937-27938	PID 4396 creates mpam-bce1d755.exe and UpdatePlatform.exe in Windows SystemTemp; two rule 92217 alerts.
13:35:14.448-.450	83314-83315; 27939	SYSTEM service logon/privilege audit and GPSvcGroup svchost.exe startup.
13:37:25.182-.386	27940-27943	Three hostname.exe and one whoami.exe launched from PowerShell PID 1392.
13:37:25.589	27944	PowerShell PID 10880 created to run Test-Path for Scenario_1_script.bat.
13:37:25.773	27945	PID 10880 creates first __PSScriptPolicyTest_*.ps1; rule 92213 alert.
13:37:26.099-.124	27946-27947	Fourth hostname.exe and second whoami.exe launched by parent PowerShell PID 1392.
13:37:26.282	27948	PowerShell PID 4500 launched with Start-Process against Scenario_1_script.bat.

UTC	Record(s)	Event / relationship
13:37:26.445	27949	PID 4500 creates second __PSScriptPolicyTest_*.ps1; rule 92213 alert.
13:37:27.087	27950	PowerShell PID 4500 spawns cmd.exe PID 6000 using /c and batch path; rule 92004 alert.
13:37:27.231	27951	OpenConsole.exe starts; no direct parent linkage to cmd.exe is recorded.
13:37:53.569	27952	Scheduled Edge updater process; no established connection to the batch.
13:39:43.850-44.639	27953-27954; 83316-83317	Windows Update client/core worker and SYSTEM service audit pair.
13:41:44.837-45.609	27955; 83318-83319	Second observed update core worker and SYSTEM service audit pair; end of supplied event span.

6. MITRE ATT&CK Mapping

Mappings below reflect **observed behavior**, independent of Wazuh rule tags. A technique association describes what the telemetry shows; it does not automatically establish adversary intent.

Tactic / technique	Supporting records	Mapping basis / limitation
Discovery - T1082, System Information Discovery	27940-27942, 27946	Four hostname.exe launches from one PowerShell parent. Local host-identification commands; outputs and ultimate purpose unknown.
Discovery - T1033, System Owner/User Discovery	27943, 27947	Two whoami.exe launches from same parent/user session. Supports user-context discovery, not credential theft.
Execution - T1059.001, PowerShell	27944, 27948	PowerShell process commands explicitly use Test-Path and Start-Process to check/launch the BAT path.
Execution - T1059.003, Windows Command Shell	27950	cmd.exe /c launched with the BAT path; precise ProcessGuid/ParentProcessGuid link to PID 4500.

Excluded alert-proposed mappings: Wazuh T1570 (Lateral Tool Transfer) for records 27937-27938 and T1105 (Ingress Tool Transfer) for 27945/27949 are **not supported** by these events. The files are locally created by known processes, without documented network source/destination or movement between systems. Do not infer credential access, persistence, privilege escalation, command-and-control, or exfiltration from this set.

7. Overall Investigation Conclusion

Observed findings: A single Windows endpoint records a concentrated sequence of four hostname checks, two whoami checks, a PowerShell batch-file existence check, and a PowerShell-initiated cmd.exe /c launch referencing Scenario_1_script.bat. Parent-child and file-creator relationships are internally consistent and corroborated by both Sysmon events and the Wazuh alert for PowerShell-to-cmd spawning. Other alerts align with Defender update staging and PowerShell-created temporary policy-test files, and are not sufficient evidence of malicious transfer.

Inferred significance: The near-synchronous repeated discovery commands, named batch path and lab-style directory naming are compatible with an intentional local test/execution sequence. The observation should be treated as **confirmed command activity with unproven malicious effect**, rather than proof of a completed intrusion. No available record reveals batch-file content, success status, downstream changes or any remote system involvement. The highest operational value is the verified process chain and careful triage of misleading alert tags.

Overall confidence: HIGH in the reconstructed process/file chronology and determination that the five alerts overlap the archived events. **Medium** confidence in the interpretation that the sequence is planned laboratory exercise activity rather than unrelated manual commands. **Insufficient evidence** to conclude what the batch script did or whether any security impact occurred. These confidence statements are restricted to this export, not a full-host forensic examination.